Mustafa Mohammed Ali

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Summary

Backend engineer with 4+ years building and running high-scale Django systems in production. I own complex modules end to end on two live B2B enterprise platforms: ticketing and SLA lifecycle, warehouse and serial-number tracking, ERP integrations, and bulk data operations. I tune queries from execution plans, design service layers, and mentor interns.

Technical Skills

- **Backend:** Python, Django, DRF, Flask, Node.js, SQL
- **Data and async:** PostgreSQL (query-plan tuning), MySQL, MongoDB, Redis, Celery, RabbitMQ, Kafka, Elasticsearch, pandas
- **APIs:** REST, WebSockets, Socket.IO, bulk and idempotent endpoints
- **Cloud and DevOps:** Docker, AWS, GCP, Nginx, CI/CD, Git
- **AI/ML:** Computer vision and NLP (YOLOv8, VGG19, PyTorch, scikit-learn, NLTK)

Experience

Holouly Innovative Solutions | Cairo, Egypt | Jul 2022 – Present

Mid-level Backend Developer | Dec 2023 – Present

- Own backend modules on ETEPS (field service management: ticketing, SLA, inventory, ERP integration) and Mdawm (attendance, live tracking, violation monitoring, CRM integration), two live B2B platforms on Django, DRF, PostgreSQL, Redis and Celery.
- Refactored the ticket-closing flow into a service-layer design with named domain exceptions and thin views.
- Built a bulk-resolve API returning HTTP 207 Multi-Status, with per-ticket atomic savepoints so one failure never rolls back the batch.
- Designed an SLA deadline engine that precomputes deadlines and updates them at write time.

- Cut ticket-list latency by reading `EXPLAIN ANALYZE` plans, adding a missing composite index and fixing `work_mem` issues.
- Fixed a serial-number corruption bug in warehouse tracking caused by deferred fields in `bulk_update`.
- Maintained the BAAN/Infor ERP integration and a SQL Server inventory sync (pyodbc, Celery); fixed a timeout that stalled sync at about 134K of 221K records, and removed SQL injection risk and N+1 queries.
- Built pandas tooling for cross-database serial matching and duplicate cleanup; developed an NLTK chatbot for data retrieval and automated reporting.
- Mentor interns: wrote a Django internship training plan and onboarding roadmap.

Junior Backend Developer | Aug 2022 – Dec 2023

- Built and optimized APIs with Django ORM, raw SQL and PostgreSQL, reducing response time by 25%.
- Contributed to backend architecture with a focus on security, scalability and maintainability.

Projects

- **Tanzim** (Django, DRF, React): multi-tenant enterprise SaaS platform; backend complete, frontend in progress.
- **Booclair** (Node.js, Flutter, Socket.IO, WebRTC): real-time chat and video meetings for book clubs. Live.
- **Dress Me Up** (graduation project): fashion recommender using YOLOv8, VGG19 and cosine similarity, with a Flask backend and Flutter app.

Education

Ain Shams University | B.Sc. Computer Science (Scientific Computing) | Sep 2020 – May 2024

- Transferred from Benha University after one year. Languages: Arabic (native), English (professional).